

Chapter 8

Security

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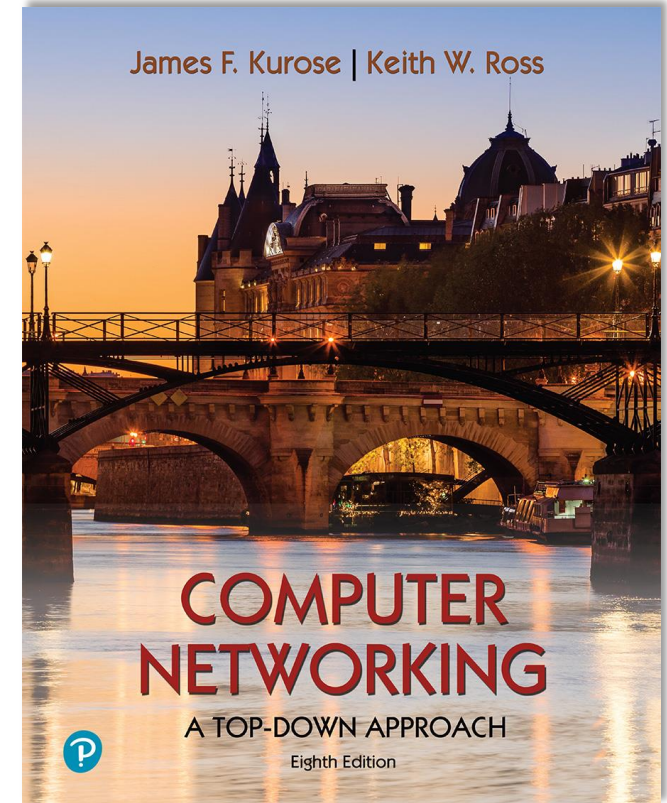
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Computer Networking: A Top-Down Approach

8th edition

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Security: overview

Chapter goals:

- understand principles of network security:
 - cryptography and its *many* uses beyond “confidentiality”
 - authentication
 - message integrity
- security in practice:
 - firewalls and intrusion detection systems
 - security in application, transport, network, link layers

Chapter 8 outline

- What is network security?
- Principles of cryptography
- Message integrity, authentication
- Securing e-mail
- Securing TCP connections: TLS
- Network layer security: IPsec
- Security in wireless and mobile networks
- Operational security: firewalls and IDS



What is network security?

confidentiality: only sender, intended receiver should “understand” message contents

- sender encrypts message
- receiver decrypts message

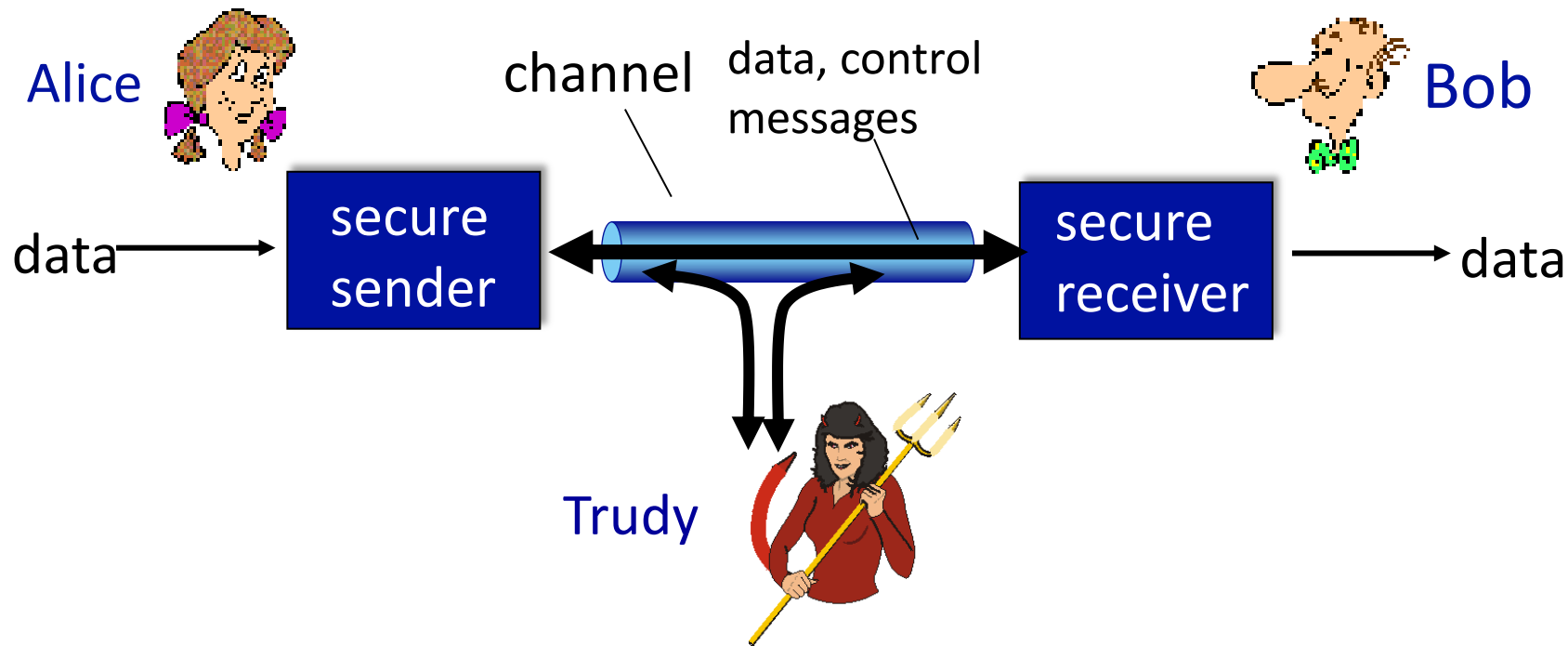
authentication: sender, receiver want to confirm identity of each other

message integrity: sender, receiver want to ensure message not altered (in transit, or afterwards) without detection

access and availability: services must be accessible and available to users

Friends and enemies: Alice, Bob, Trudy

- well-known in network security world
- Bob, Alice (lovers!) want to communicate “securely”
- Trudy (intruder) may intercept, delete, add messages



Friends and enemies: Alice, Bob, Trudy

Who might Bob and Alice be?

- ... well, *real-life* Bobs and Alices!
- Web browser/server for electronic transactions (e.g., on-line purchases)
- on-line banking client/server
- DNS servers
- BGP routers exchanging routing table updates
- other examples?

There are bad guys (and girls) out there!

Q: What can a “bad guy” do?

A: A lot! (recall section 1.6)

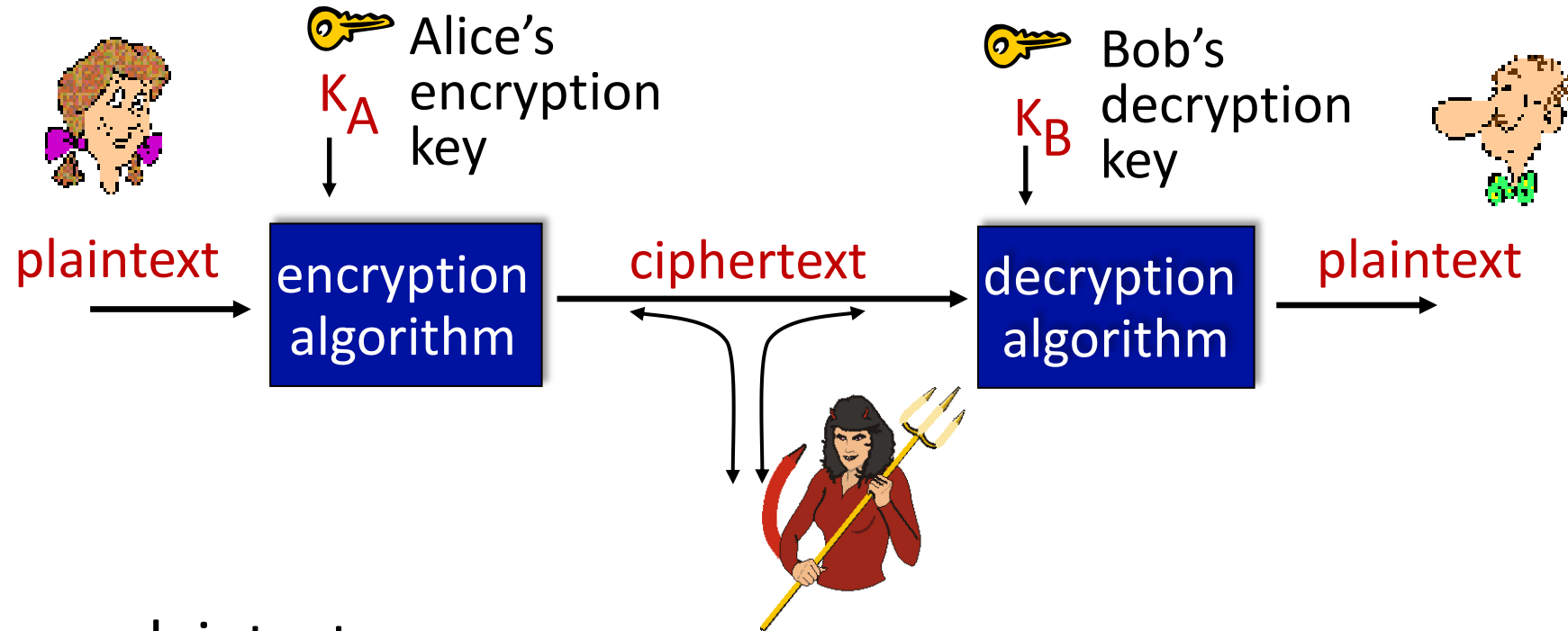
- **eavesdrop**: intercept messages
- actively **insert** messages into connection
- **impersonation**: can fake (spoof) source address in packet (or any field in packet)
- **hijacking**: “take over” ongoing connection by removing sender or receiver, inserting himself in place
- **denial of service**: prevent service from being used by others (e.g., by overloading resources)

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The language of cryptography



m : plaintext message

$K_A(m)$: ciphertext, encrypted with key K_A

$m = K_B(K_A(m))$

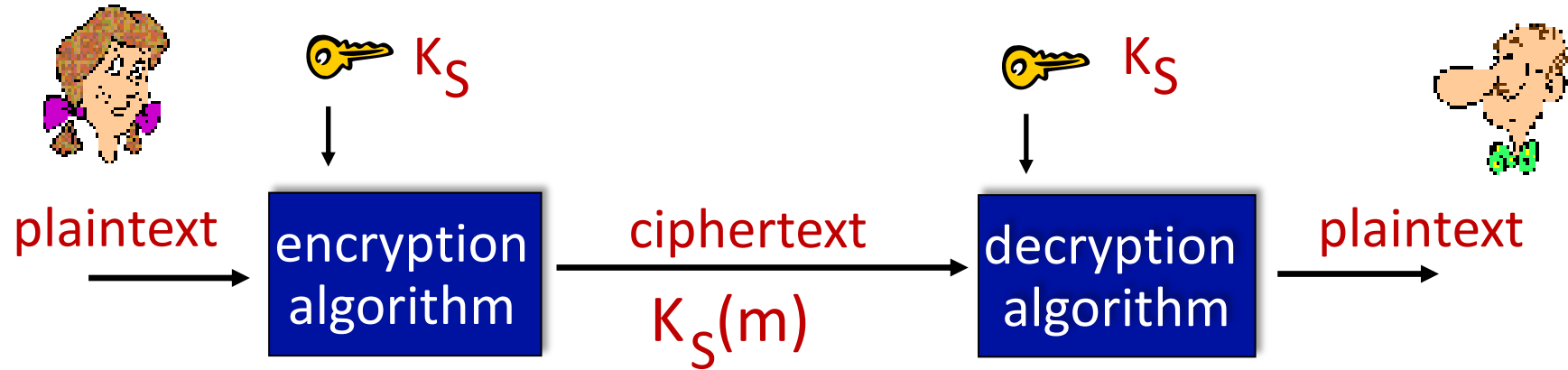
Breaking an encryption scheme

- **cipher-text only attack:**
Trudy has ciphertext she can analyze

- **two approaches:**
 - brute force: search through all keys
 - statistical analysis

- **known-plaintext attack:**
Trudy has plaintext corresponding to ciphertext
 - *e.g.*, in monoalphabetic cipher, Trudy determines pairings for a,l,i,c,e,b,o,
- **chosen-plaintext attack:**
Trudy can get ciphertext for chosen plaintext

Symmetric key cryptography



symmetric key crypto: Bob and Alice share same (symmetric) key: K

- e.g., key is knowing substitution pattern in mono alphabetic substitution cipher

Q: how do Bob and Alice agree on key value?

Simple encryption scheme

substitution cipher: substituting one thing for another

- monoalphabetic cipher: substitute one letter for another

plaintext:	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
		↓																								↓
ciphertext:	m	n	b	v	c	x	z	a	s	d	f	g	h	j	k	l	p	o	i	u	y	t	r	e	w	q

e.g.: Plaintext: bob. i love you. alice
ciphertext: nkn. s gktc wky. mgsbc

🔑 *Encryption key*: mapping from set of 26 letters
to set of 26 letters

A more sophisticated encryption approach

- n substitution ciphers, $M_1, M_2, \dots, M_n$
- cycling pattern:
 - e.g., $n=4$: M_1, M_3, M_4, M_3, M_2 ; M_1, M_3, M_4, M_3, M_2 ; ..
- for each new plaintext symbol, use subsequent substitution pattern in cyclic pattern
 - dog: d from M_1 , o from M_3 , g from M_4
- 🔑 **Encryption key:** n substitution ciphers, and cyclic pattern
 - key need not be just n-bit pattern

Symmetric key crypto: DES

DES: Data Encryption Standard

- US encryption standard [NIST 1993]
- 56-bit symmetric key, 64-bit plaintext input
- block cipher with cipher block chaining
- how secure is DES?
 - DES Challenge: 56-bit-key-encrypted phrase decrypted (brute force) in less than a day
 - no known good analytic attack
- making DES more secure:
 - 3DES: encrypt 3 times with 3 different keys

AES: Advanced Encryption Standard

- symmetric-key NIST standard, replaced DES (Nov 2001)
- processes data in 128 bit blocks
- 128, 192, or 256 bit keys
- brute force decryption (try each key) taking 1 sec on DES, takes 149 trillion years for AES

Public Key Cryptography

symmetric key crypto:

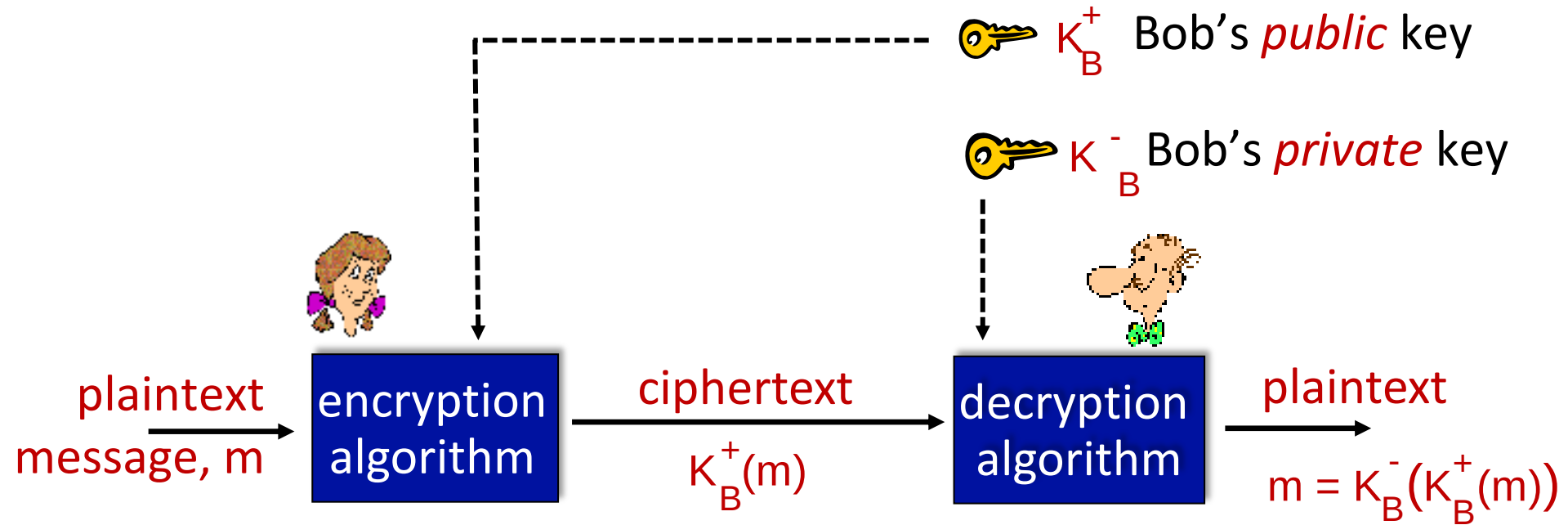
- requires sender, receiver know shared secret key
- Q: how to agree on key in first place (particularly if never “met”)?

public key crypto

- *radically* different approach [Diffie-Hellman76, RSA78]
- sender, receiver do *not* share secret key
- *public* encryption key known to *all*
- *private* decryption key known only to receiver



Public Key Cryptography



Wow - public key cryptography revolutionized 2000-year-old (previously only symmetric key) cryptography!

- similar ideas emerged at roughly same time, independently in US and UK (classified)

Public key encryption algorithms

requirements:

① need $K_B^+(\cdot)$ and $K_B^-(\cdot)$ such that

$$K_B^-(K_B^+(m)) = m$$

② given public key K_B^+ , it should be impossible to compute private key K_B^-

RSA: Rivest, Shamir, Adelson algorithm

Prerequisite: modular arithmetic

- $x \bmod n$ = remainder of x when divide by n

- facts:

$$[(a \bmod n) + (b \bmod n)] \bmod n = (a+b) \bmod n$$

$$[(a \bmod n) - (b \bmod n)] \bmod n = (a-b) \bmod n$$

$$[(a \bmod n) * (b \bmod n)] \bmod n = (a*b) \bmod n$$

- thus

$$(a \bmod n)^d \bmod n = a^d \bmod n$$

- example: $x=14$, $n=10$, $d=2$:

$$(x \bmod n)^d \bmod n = 4^2 \bmod 10 = 6$$

$$x^d = 14^2 = 196 \quad x^d \bmod 10 = 6$$

RSA: getting ready

- message: just a bit pattern
- bit pattern can be uniquely represented by an integer number
- thus, encrypting a message is equivalent to encrypting a number

example:

- $m = 10010001$. This message is uniquely represented by the decimal number 145.
- to encrypt m , we encrypt the corresponding number, which gives a new number (the ciphertext).

RSA: Creating public/private key pair

1. choose two large prime numbers p, q . (e.g., 1024 bits each)
2. compute $n = pq$, $z = (p-1)(q-1)$
3. choose e (with $e < n$) that has no common factors with z (e, z are “relatively prime”).
4. choose d such that $ed-1$ is exactly divisible by z . (in other words: $ed \bmod z = 1$).
5. *public* key is (n, e) . *private* key is (n, d) .
 $\underbrace{(n, e)}_{K_B^+}$ $\underbrace{(n, d)}_{K_B^-}$

RSA: encryption, decryption

0. given (n, e) and (n, d) as computed above
1. to encrypt message $m (< n)$, compute
$$c = m^e \bmod n$$
2. to decrypt received bit pattern, c , compute
$$m = c^d \bmod n$$

magic happens!

$$m = \underbrace{(m^e \bmod n)}_c^d \bmod n$$

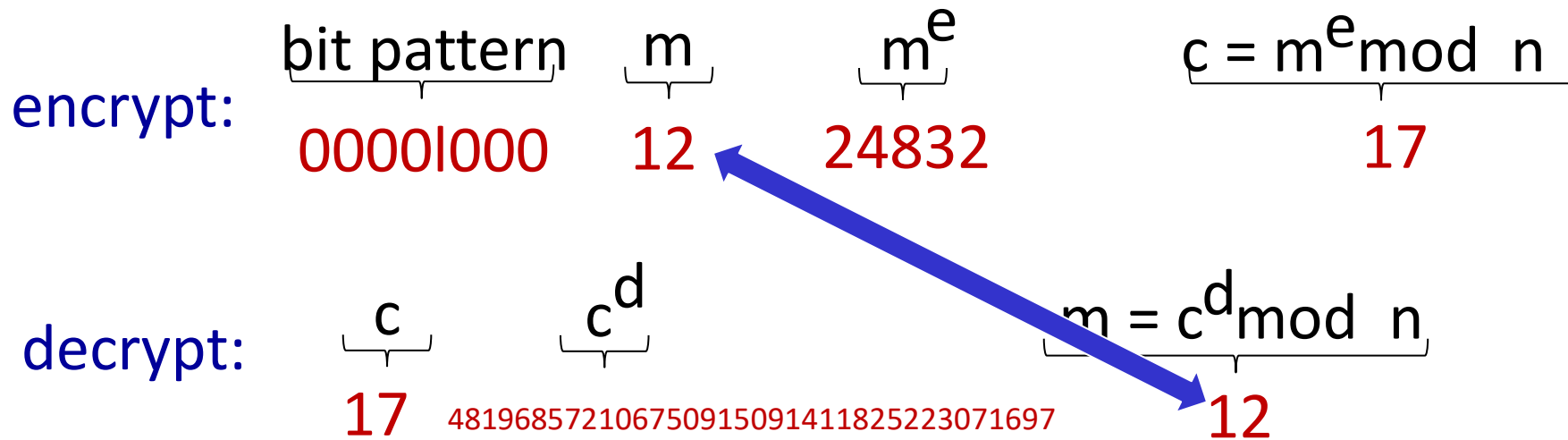
RSA example:

Bob chooses $p=5$, $q=7$. Then $n=35$, $z=24$.

$e=5$ (so e , z relatively prime).

$d=29$ (so $ed-1$ exactly divisible by z).

encrypting 8-bit messages.



Why does RSA work?

- must show that $c^d \bmod n = m$, where $c = m^e \bmod n$
- fact: for any x and y : $x^y \bmod n = x^{(y \bmod z)} \bmod n$
 - where $n = pq$ and $z = (p-1)(q-1)$
- thus,
$$\begin{aligned}c^d \bmod n &= (m^e \bmod n)^d \bmod n \\&= m^{ed} \bmod n \\&= m^{(ed \bmod z)} \bmod n \\&= m^1 \bmod n \\&= m\end{aligned}$$

RSA: another important property

The following property will be *very* useful later:

$$\underbrace{K_B^-(K_B^+(m))}_{\text{use public key first, followed by private key}} = m = \underbrace{K_B^+(K_B^-(m))}_{\text{use private key first, followed by public key}}$$

use public key
first, followed
by private key

use private key
first, followed
by public key

result is the same!

Why $K_B^-(K_B^+(m)) = m = K_B^+(K_B^-(m))$?

follows directly from modular arithmetic:

$$\begin{aligned}(m^e \bmod n)^d \bmod n &= m^{ed} \bmod n \\ &= m^{de} \bmod n \\ &= (m^d \bmod n)^e \bmod n\end{aligned}$$

Why is RSA secure?

- suppose you know Bob's public key (n,e) . How hard is it to determine d ?
- essentially need to find factors of n without knowing the two factors p and q
 - fact: factoring a big number is hard

RSA in practice: session keys

- exponentiation in RSA is computationally intensive
- DES is at least 100 times faster than RSA
- use public key crypto to establish secure connection, then establish second key – symmetric session key – for encrypting data

session key, K_s

- Bob and Alice use RSA to exchange a symmetric session key K_s
- once both have K_s , they use symmetric key cryptography